

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$.)

```
KD2_Pritam_99277>DELIMITER $$
KD2_Pritam_99277>
KD2_Pritam_99277>CREATE PROCEDURE sp_rectangle()
-> BEGIN
->     DECLARE v_length INT DEFAULT 10;
->     DECLARE v_width INT DEFAULT 5;
->     DECLARE v_perimeter INT;
->     DECLARE v_area INT;
->
->     SET v_perimeter = 2 * (v_length + v_width);
->     SET v_area = v_length * v_width;
->
->     SELECT v_length AS Length,
->            v_width AS Width,
->            v_perimeter AS Perimeter,
->            v_area AS Area;
-> END;
-> $$
Query OK, 0 rows affected (0.07 sec)

KD2_Pritam_99277>
KD2_Pritam_99277>DELIMITER ;
KD2_Pritam_99277>
KD2_Pritam_99277>CALL sp_rectangle();
+-----+-----+-----+-----+
| Length | Width | Perimeter | Area |
+-----+-----+-----+-----+
|      10 |      5 |          30 |      50 |
+-----+-----+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```

KD2_Pritam_99277>CREATE TABLE temp (
->   num INT,
->   square INT,
->   cube1 INT
-> );
Query OK, 0 rows affected (0.05 sec)

KD2_Pritam_99277>SELECT * FROM temp;
Empty set (0.01 sec)

KD2_Pritam_99277>DELIMITER $$
KD2_Pritam_99277>
KD2_Pritam_99277>CREATE PROCEDURE sp_number()
-> BEGIN
->   DECLARE num INT DEFAULT 5;
->
->   INSERT INTO temp
->   VALUES (num, num * num, num * num * num);
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_number already exists
KD2_Pritam_99277>
KD2_Pritam_99277>DELIMITER ;
KD2_Pritam_99277>
KD2_Pritam_99277>CALL sp_number();
Query OK, 1 row affected (0.01 sec)

KD2_Pritam_99277>
KD2_Pritam_99277>SELECT * FROM temp;
+-----+-----+-----+
| num | square | cube1 |
+-----+-----+-----+
| 5 | 25 | 125 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```

KD2_Pritam_99277>DELIMITER $$
KD2_Pritam_99277>
KD2_Pritam_99277>CREATE PROCEDURE sp_fahrenheit_to_celsius(p_fahrenheit DECIMAL(10,2))
-> BEGIN
->   DECLARE v_celsius DECIMAL(10,2);
->
->   SET v_celsius = (p_fahrenheit - 32) * 5 / 9;
->
->   SELECT p_fahrenheit AS Fahrenheit,
->          v_celsius AS Celsius;
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Pritam_99277>
KD2_Pritam_99277>DELIMITER ;
KD2_Pritam_99277>
KD2_Pritam_99277>CALL sp_fahrenheit_to_celsius(100);
+-----+-----+
| Fahrenheit | Celsius |
+-----+-----+
| 100.00 | 37.78 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```

```

KD2_Pritam_99277>DELIMITER $$
KD2_Pritam_99277>
KD2_Pritam_99277>CREATE PROCEDURE sp_celsius_to_fahrenheit(p_celsius DECIMAL(10,2))
-> BEGIN
->     DECLARE v_fahrenheit DECIMAL(10,2);
->
->     SET v_fahrenheit = (9 * p_celsius / 5) + 32;
->
->     SELECT p_celsius AS Celsius,
->            v_fahrenheit AS Fahrenheit;
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Pritam_99277>
KD2_Pritam_99277>DELIMITER ;
KD2_Pritam_99277>
KD2_Pritam_99277>CALL sp_celsius_to_fahrenheit(37.78);
+-----+-----+
| Celsius | Fahrenheit |
+-----+-----+
| 37.78 | 100.00 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```

KD2_Pritam_99277>DELIMITER $$
KD2_Pritam_99277>
KD2_Pritam_99277>CREATE PROCEDURE sp_convert_inches(p_inches INT)
-> BEGIN
->     DECLARE v_yards INT;
->     DECLARE v_feet INT;
->     DECLARE v_inches INT;
->
->     SET v_yards = p_inches DIV 36;
->     SET v_feet = (p_inches MOD 36) DIV 12;
->     SET v_inches = p_inches MOD 12;
->
->     SELECT p_inches AS Total_Inches,
->            v_yards AS Yards,
->            v_feet AS Feet,
->            v_inches AS Inches;
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

KD2_Pritam_99277>
KD2_Pritam_99277>DELIMITER ;
KD2_Pritam_99277>
KD2_Pritam_99277>CALL sp_convert_inches(124);
+-----+-----+-----+-----+
| Total_Inches | Yards | Feet | Inches |
+-----+-----+-----+-----+
|          124 |      3 |    1 |      4 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```

KD2_Pritam_99277>CREATE PROCEDURE sp_check_product(
->    p_num1 DECIMAL(10,2),
->    p_num2 DECIMAL(10,2)
-> )
-> BEGIN
->     DECLARE v_product DECIMAL(10,2);
->
->     SET v_product = p_num1 * p_num2;
->
->     IF v_product >= 100 THEN
->         SELECT p_num1 AS Number1,
->                p_num2 AS Number2,
->                v_product AS Product,
->                'Product is equal to or greater than 100' AS Result;
->     ELSE
->         SELECT p_num1 AS Number1,
->                p_num2 AS Number2,
->                v_product AS Product,
->                'Product is less than 100' AS Result;
->     END IF;
-> END;
-> $$

```

Query OK, 0 rows affected (0.01 sec)

```

KD2_Pritam_99277>

```

```

KD2_Pritam_99277>DELIMITER ;

```

```

KD2_Pritam_99277>

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```

KD2_Pritam_99277>CALL sp_check_product(10.5, 10);

```

Number1	Number2	Product	Result
10.50	10.00	105.00	Product is equal to or greater than 100

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)